MILITARY SPECIFICATION SHEET

BIT, SCREWDRIVER, HI-TORQUE, TYPE V

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the screwdriving bits described herein shall consist of this document and the latest issue of specification MIL-B-9946.

1. SCOPE

1.1 Scope. The tool assembly applicable herein shall be a male hexagon driven screwdriver tool with a wheel shaped bit for engaging and driving fasteners equipped with driving recesses conforming to MS 33750.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-T-570	Tool Steel, Alloy
GGG-W-641	Wrench, Socket; (and Sockets, Handles, and Attachments for Socket Wrenches; Hand)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specification and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Beneficial comments (recommendations, additions, delections) and any pertinent data which may be of use in improving this document should be addressed to AFLC CASO/LODS, Federal Center, Battle Creek, Mich 49016 by using the self-addressed Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

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(Single copies of Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA).

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Ferederal Specifications and Standards from established distribution points in their agencies.)

STANDARDS

Military

MS33750	Recess - Hi-Torque, Dimensions of Recess Gage, and Driver for

National

NAS 583 Thru 590	Bolt, 100° Flush Tension Head, "Hi-Torque" Recess, Alloy Steel, 160,000 PSI
NAS 1243 Thru 1250	Bolt, 100° Flush Tension Head, "Hi-Torque" Recess, .0156 Oversize Shank, Short Thread, Alloy Steel, 160,000 PSI

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) INC., STANDARDS:

B107.4 Driving and Spindle Ends for Portable Hand, Air and Electric Tools

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Design. The tool shall be an assembly of a male hexagon plug driven member of a Rockwell "C" scale hardness of 45 to 55, conforming to ANSI B 107.4 and GGG-W-641, with an extension on the opposite end from the hexagon plug fitted with a circular wheel shaped part for driving screws. For further reference the hexagon plug shaped part will be referred to as the

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"adapter" and the wheel shaped part will be the "driving blade". This driving blade shall be of the required configuration of driving screw-head fasteners as applicable to MS 33750. The assembly shall have provisions for adjusting the driving blade to several peripheral positions and shall withstand the torque testing requirements as specified herein.

3.2 Material. QQ-T-570, Type S2 (or equal).

3.3 Coatings. Rust preventive, for interim storage.

3.4 Configuration. Details of configuration as shown on Figure 1 and Table I shall be complied with.

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TABLE I - DETAILS OF TOOL ASSEMBLY

TOOL ASSEMBLY DESIGNATION	DRIVING BLADE DESIGNATION	NOM. HEXAGON DRIVE SIZE (IN.) "A"	NOM. DRIVING BLADE SIZE (IN.) "B"	"C" +.010 -.000 (IN.)	MIN. TOTAL LENGTH (IN.) "D"	"E" (IN.) MAX./ MIN.	"F" RAD. (IN) ±.010	APPLICABLE BOLT SIZE (IN)
HT2504M	HTB4M	5/16	7/8	.059	7/8	.0740 .0710	.095	1/4
HT2505M	HTB5M	5/16	1	.069	1	.0800 .0770	.095	5/16
HT2506M	HTB6M	5/16	1 1/4	.093	1 1/4	.0960 .0930	.095	3/8
HT2507M	HTB7M	5/16	1 3/8	.106	1 3/8	.1130 .1095	.182	7/16
HT4403M	HTB3M	1/4	9/16	.045	9/16	.0600 .0575	.095	3/16
HT4404M	HTB4M	1/4	7/8	.059	7/8	.0740 .0710	.095	1/4
HT4405M	HTB5M	1/4	1	.069	1	.0800 .0770	.095	5/16
HT4703M	HTB3M	7/16	9/16	.045	9/16	.0600 .0575	.095	3/16
HT4704M	HTB4M	7/16	7/8	.059	7/8	.0740 .0710	.095	1/4
HT4705M	HTB5M	7/16	1	.069	1	.0800 .0770	.095	5/16

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4. QUALITY ASSURANCE PROVISIONS

4.1 Test block. The test block to be used for sample testing shall be manufactured by the supplier with a testing slot in accordance with the applicable screw driving slots as shown on NAS 583 through NAS 590 or NAS 1243 through 1250 with the exception that the testing slot shall accommodate from 57 to 62 degrees angular portion of the driving blade being tested. A comparison of driving blades to applicable NAS numbers and minimum torque testing requirements are shown in Table II.

4.2 Tool testing. Sample testing of driving blades shall withstand torque forces shown in Table II without any indication of failure.

TABLE II - NAS COMPARISON AND TORQUE TESTING REQUIREMENTS

Blade DESIGNATION	NAS	Min. Torque TESTING REQUIREMENT (In.-Pounds)
HTB3M	583 1243	100
HTB4M	584 1244	250
HTB5M	585 1245	500
HTB6M	586 1246	860
HTB7M	587 1247	1850

5. PREPARATION FOR DELIVERY

(This section is not applicable. It is covered in MIL-B-9946).

6. NOTES

6.1 Ordering data. Identification of item required as applicable to paragraph 6.2 of MIL-B-9946 should show the part number and nomenclature required, as referenced in column 1 or 2 of Table I.

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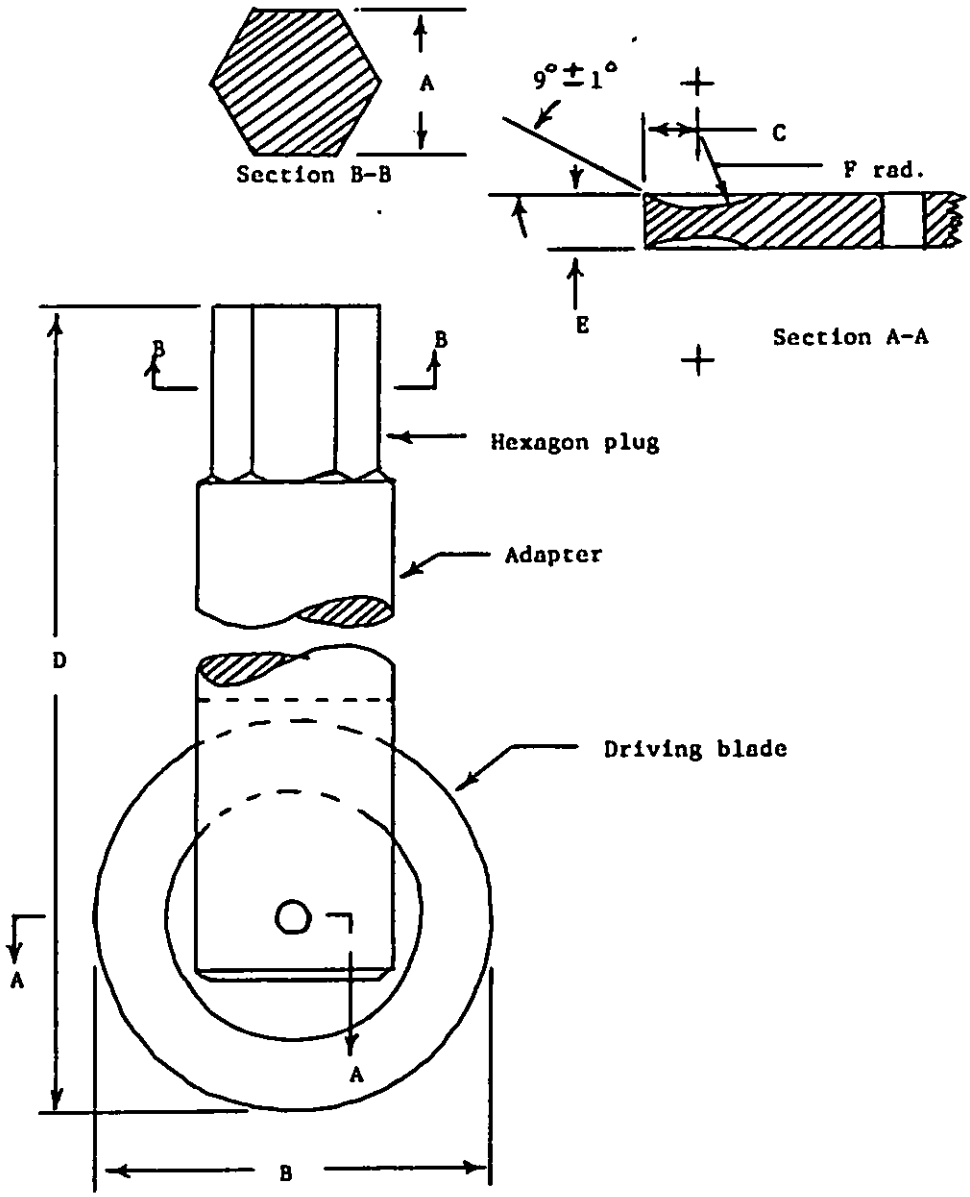
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Tool Assembly
Figure 1

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